



Topic 7: Multiple Access

CSE 4405: Data and Telecommunications

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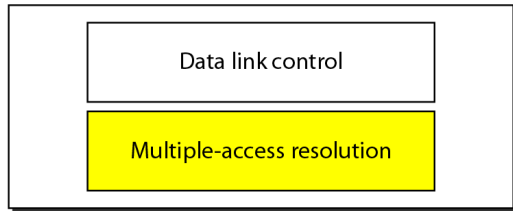
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Islamic University of Technology (IUT)

Data Link Layer and Shared-Media Access

- Data link functionality is often viewed as two sublayers.
- **Upper sublayer:** data-link control tasks.
- **Lower sublayer:** resolves who can use a shared medium.
- Multiple-access protocols belong to this lower access-control role.

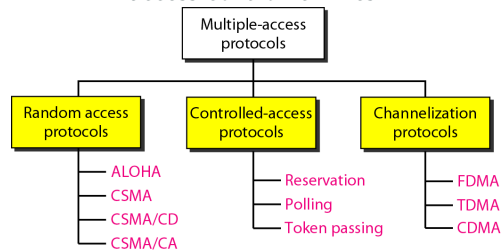
Figure: Separates control functions from shared-medium access arbitration.
Data link layer



Taxonomy of Multiple-Access Protocols

- When many stations use one broadcast or multipoint link, coordination is required.
- A multiple-access protocol determines how stations share transmission opportunities.
- This chapter organizes methods into random access, controlled access, and channelization.

Figure: Presents the chapter roadmap of access-control families.



12-1 Random Access: Scope

- In contention-based access, no station is permanently superior to others.
- A station with data decides independently whether to transmit, following protocol rules.
- Collisions can occur; each method defines how to reduce and recover from them.

Topics in this section

- ALOHA
- Carrier Sense Multiple Access (CSMA)
- CSMA with Collision Detection (CSMA/CD)
- CSMA with Collision Avoidance (CSMA/CA)

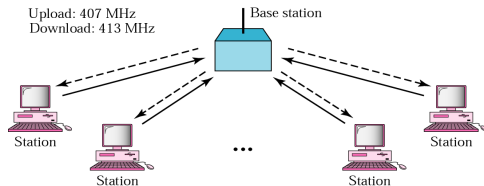
Random Access: Core Questions

- Every station has the right to attempt transmission.
- If two or more stations transmit together, frames collide and become unusable.
- To operate correctly, each station follows a decision procedure:
 - When may I transmit?
 - What if the channel is busy?
 - How do I know success or failure?
 - What do I do after collision?

ALOHA Network: Earliest Random Access

- Developed at the University of Hawaii in the early 1970s.
- Uses a central base station as the communication hop.
- Multiple users share the same incoming channel, so collisions are possible.
- It established foundational ideas for later random-access protocols.

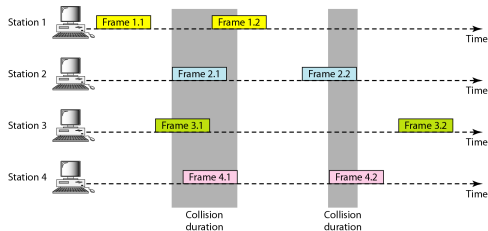
Figure: An early wireless shared-channel architecture where simultaneous attempts can conflict.



Pure ALOHA

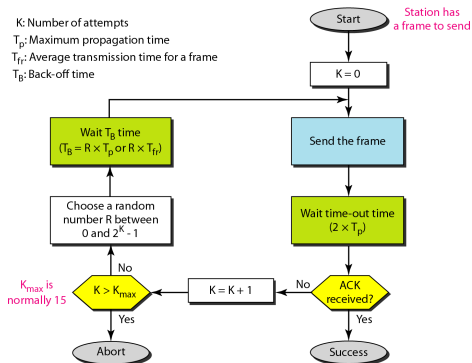
- A station sends whenever it has a frame.
- One common channel is shared by all stations.
- If two frames overlap in time, collision occurs.
- Reliability comes from retransmission after failure.

Figure: Illustrates unslotted transmissions and possible frame overlap.



Procedure for Pure ALOHA

Figure: After transmission, the station waits for feedback; on failure, it backs off randomly and retries.



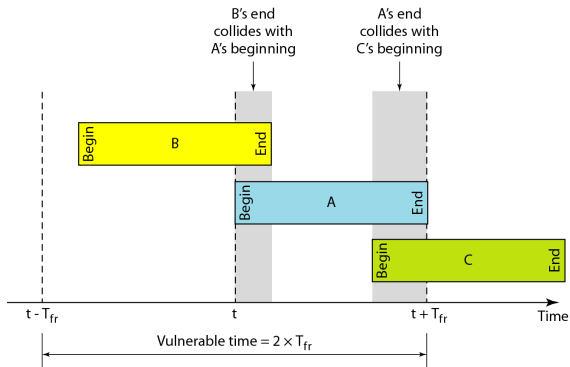
ALOHA Protocol Rule: ACK and Backoff

- After sending, the station waits for an acknowledgment.
- If no ACK arrives within roughly $2T_p$, it assumes loss due to collision.
- Retransmission is delayed by a random backoff T_B , commonly exponential.
- Typical form: $T_B = R \times T_p$ or $T_B = R \times T_{fr}$.
- In binary exponential backoff, $R \in [0, 2^k - 1]$ after k failed attempts.

T_p : maximum propagation time T_{fr} : frame transmission time

Vulnerable Time in Pure ALOHA

Figure: Any overlapping transmission inside a $2T_{fr}$ window can destroy the frame.



Pure ALOHA Throughput

Throughput model: $S = Ge^{-2G}$

Maximum throughput: $S_{\max} = 0.184$ when $G = \frac{1}{2}$

- S : average number of successful transmissions per frame time.
- G : offered load (attempts per T_{fr}); it is **not** a time quantity.
- If attempt rate is λ attempts/s, then $G = \lambda T_{fr}$ (dimensionless).
- Peak efficiency is about 18.4%, so Pure ALOHA is simple but bandwidth-inefficient.

Pure ALOHA: Step-by-Step Derivation

- Assume there are n stations. In one T_{fr} , each station attempts with probability p .
- So expected attempts in one T_{fr} are $G = np$. This is why G is “attempts per frame time”.
- A tagged frame survives only if no other station starts in the vulnerable window $2T_{fr}$.
- For one other station: “silent in previous T_{fr} ” and “silent in next T_{fr} ” gives $(1 - p)^2$.
- For all other $n - 1$ stations (independent attempts):

$$P_{\text{succ}|\text{tagged}} = (1 - p)^{2(n-1)}.$$

- Hence throughput (successful frames per T_{fr}):

$$S_n = np(1 - p)^{2(n-1)}.$$

Pure ALOHA: Step-by-Step Derivation

- Keep $G = np$ fixed and let $n \rightarrow \infty$ ($p = G/n$); this keeps total offered load unchanged while modeling many contending stations:

$$S_n = G \left(1 - \frac{G}{n}\right)^{2(n-1)} \longrightarrow Ge^{-2G}$$

- because $\lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n = e^{-x}$.

- Therefore $S = Ge^{-2G}$. Maximize:

$$\frac{dS}{dG} = e^{-2G}(1 - 2G) = 0 \Rightarrow G^* = \frac{1}{2},$$

$$S_{\max} = S\left(\frac{1}{2}\right) = \frac{1}{2e} \approx 0.184.$$

Pure ALOHA: Finite- n Optimum Before Large- n Limit

- Optimize for fixed n :

$$\ln S_n = \ln n + \ln p + 2(n-1) \ln(1-p),$$
$$\frac{d}{dp} \ln S_n = \frac{1}{p} - \frac{2(n-1)}{1-p} = 0 \Rightarrow p^* = \frac{1}{2n-1}.$$

- Therefore

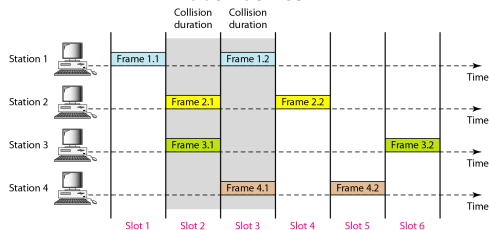
$$S_{n,\max} = \frac{n}{2n-1} \left(\frac{2n-2}{2n-1} \right)^{2n-2}, \quad \lim_{n \rightarrow \infty} S_{n,\max} = \frac{1}{2e}.$$

- Reason for this limit: it gives a clean throughput law in terms of G only (independent of a specific n, p pair).
- Using $G = np$, the large- n form becomes $S = Ge^{-2G}$, consistent with the previous frame.

Slotted ALOHA

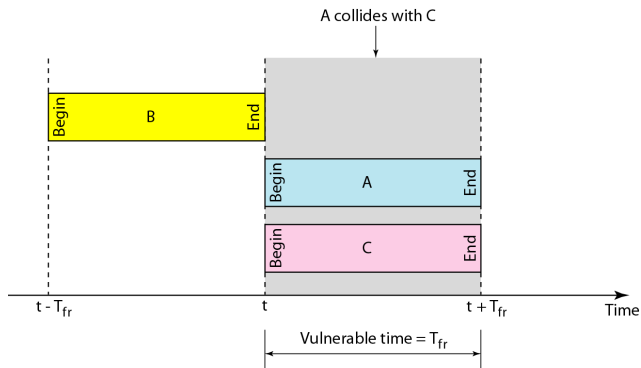
- Time is divided into slots of duration T_{fr} .
- Stations may transmit only at slot boundaries.
- This prevents partial overlaps and reduces collision window.
- Synchronization is required among participating stations.

Figure: Shows frame starts aligned to slot boundaries.



Vulnerable Time in Slotted ALOHA

Figure: The vulnerable period becomes T_{fr} , which improves success probability.



Slotted ALOHA Throughput

Throughput model: $S = Ge^{-G}$

Maximum throughput: $S_{\max} = 0.368$ when $G = 1$

- Here $G = np$: expected attempts in one slot (slot duration is T_{fr}).
- Slotting roughly doubles best-case throughput compared with Pure ALOHA.
- Peak efficiency reaches about 36.8%.

Slotted ALOHA: Step-by-Step Derivation

- Assume n backlogged stations, each transmits in a slot with probability p .
- A successful slot needs exactly one transmitter:

$$S = np(1 - p)^{n-1}, \quad G = np.$$

- Reason: choose which one transmits (n ways), that station transmits (p), all others stay silent ($(1 - p)^{n-1}$).
- Keep $G = np$ fixed and let $n \rightarrow \infty$ ($p = G/n$); this preserves slot load while moving to a many-user regime:

$$S_n = G \left(1 - \frac{G}{n}\right)^{n-1} \longrightarrow Ge^{-G}.$$

- So the throughput model is $S = Ge^{-G}$.

Slotted ALOHA: Step-by-Step Derivation

- Start from finite- n throughput:

$$S_n = np(1-p)^{n-1}.$$

- For fixed n , maximize S by

$$\frac{d}{dp} [np(1-p)^{n-1}] = 0 \Rightarrow p^* = \frac{1}{n}.$$

- Hence

$$S_{\max,n} = \left(1 - \frac{1}{n}\right)^{n-1}, \quad \lim_{n \rightarrow \infty} S_{\max,n} = \frac{1}{e} = 0.368.$$

- Reason for the limit: it expresses performance with one load variable G , which is easier to use in analysis and graphs.
- In G -form:

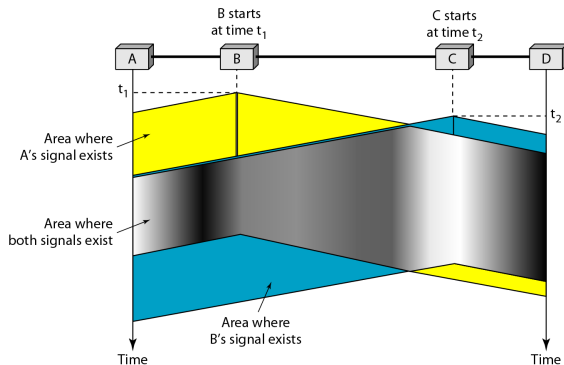
$$\frac{d}{dG} (Ge^{-G}) = e^{-G}(1-G) = 0 \Rightarrow G^* = 1, \quad S_{\max} = 1/e.$$

Carrier Sense Multiple Access (CSMA)

- CSMA follows “sense before transmit” (listen before talk).
- By checking channel state first, many avoidable collisions are prevented.
- Collisions still cannot be fully eliminated.
- Reason: propagation delay may hide in-flight transmissions from distant stations.

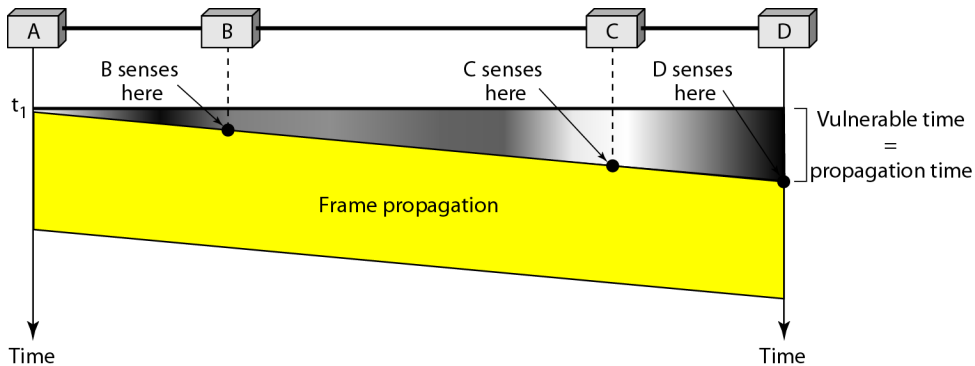
CSMA Collision Space/Time Model

Figure: Two stations can both sense idle and start before each hears the other's first bit.



Vulnerable Time in CSMA

Figure: Propagation delay T_p determines the residual collision window in carrier-sensing systems.



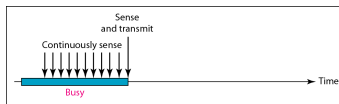
Persistence Methods

- Persistence strategy defines behavior after sensing a busy medium.
- **1-persistent:** transmit immediately once channel is idle.
- **Non-persistent:** wait random time, then sense again.
- **p -persistent:** in slotted operation, transmit with probability p ; defer with $1 - p$.

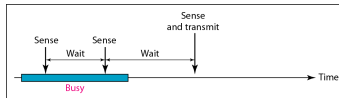
Method	Typical tradeoff
1-persistent	Lower waiting, but higher chance of simultaneous starts
Non-persistent	Fewer collisions, but more idle waiting overhead
p -persistent	Tunable compromise between aggressiveness and caution

Behavior of Persistence Methods

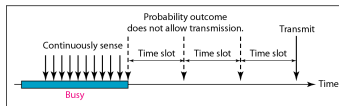
Figure: Different persistence choices reshape delay and collision probability.



a. 1-persistent



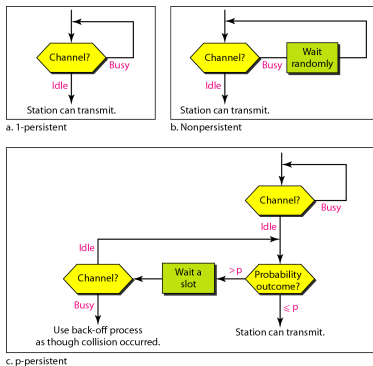
b. Nonpersistent



c. p-persistent

Flow Diagram of Persistence Methods

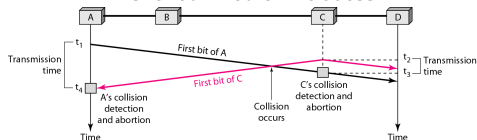
Figure: Carrier sensing plus persistence rules create the station's transmit/defer decision loop.



CSMA with Collision Detection (CSMA/CD)

- Adds collision-handling behavior to CSMA.
- A station monitors the medium while transmitting.
- If collision is detected, it aborts and retries later.
- This was central to classic shared-medium Ethernet.

Figure: Shows detection of first-bit collision in shared-medium access.



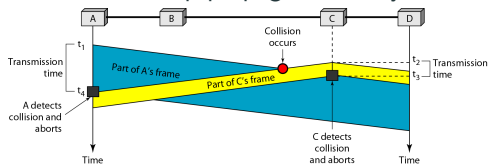
CSMA/CD: Minimum Frame Size Constraint

- For collision detection to work, transmission must last long enough.
- Required condition:

$$T_{fr} \geq 2T_p$$

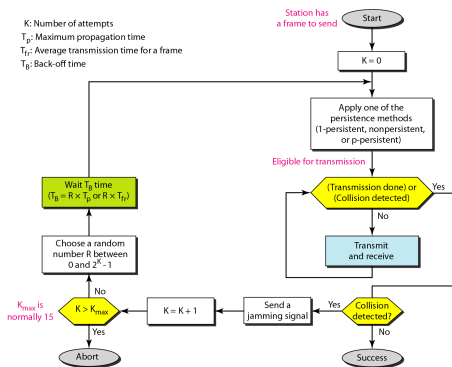
- Otherwise, a sender might finish before collision evidence returns.

Figure: Relates collision detection to round-trip propagation delay.



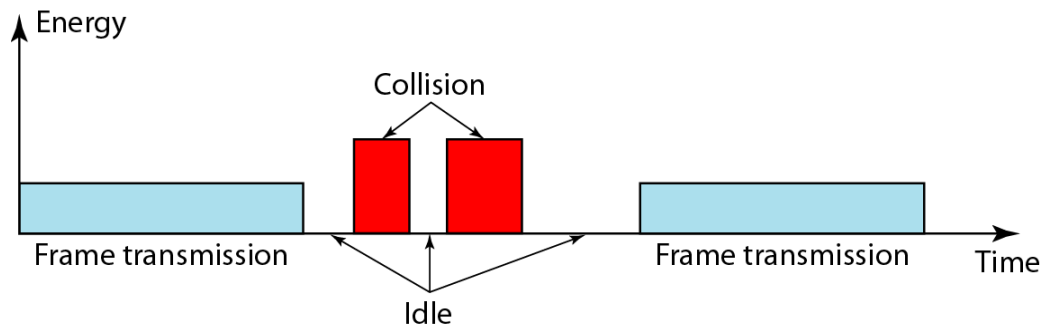
CSMA/CD Flow Diagram

Figure: Sense, transmit, detect, abort, back off, and retry form the CSMA/CD control cycle.



Energy Levels in CSMA/CD

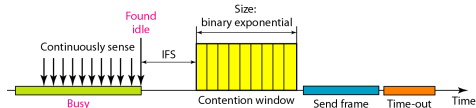
Figure: Collision produces a distinguishable energy pattern used for detection.



CSMA with Collision Avoidance (CSMA/CA)

- Designed for wireless links where collision detection is difficult.
- Avoids collisions using three mechanisms:
 - Interframe Space (IFS)
 - Contention Window (random backoff)
 - Acknowledgments (ACKs)
- The goal is prevention before data overlap occurs.

Figure: Illustrates CSMA/CA timing with IFS and random backoff.



Note

In CSMA/CA, IFS duration can be used to encode priority. Frames/stations with shorter waiting rules may access the channel sooner.



CSMA/CA Note: Busy Channel During Backoff

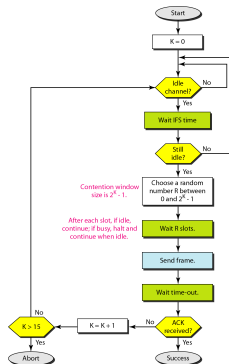
Note

If the channel becomes busy, a station **pauses** the contention timer; it does not restart from the beginning. Countdown resumes when the channel is idle again.



CSMA/CA Flow Diagram

Figure: Carrier sensing, timer freezing/resuming, and ACK-based retry drive wireless contention control.



12-2 Controlled Access: Scope

- Stations consult one another to decide who may transmit.
- A station transmits only when authorized by protocol rules.
- This reduces collisions at the cost of coordination overhead.

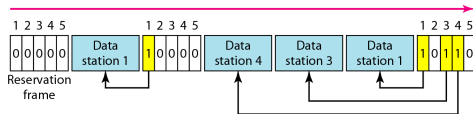
Topics in this section

- Reservation
- Polling
- Token Passing

Reservation Access Method

- Each interval begins with a reservation frame.
- With N stations, reservation frame has N minislots.
- Station i reserves by marking its own minislot.
- Reserved stations transmit data after reservation phase.

Figure: Separates contention-free reservations from the following data phase.

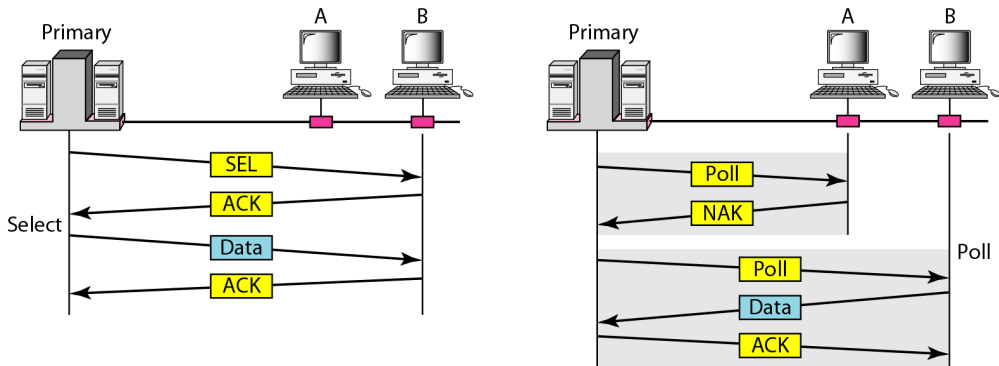


Polling Access Method

- One station is designated **primary**; others are **secondary**.
- The primary controls link operation and sequencing.
- **Poll function:** primary asks secondaries whether they have data to send.
- **Select function:** primary instructs a secondary to prepare to receive.
- Secondary stations follow primary control messages.

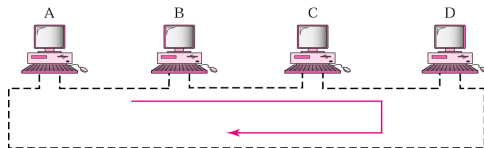
Select and Poll Functions

Figure: Primary station alternates between soliciting uplink data and assigning downlink reception.



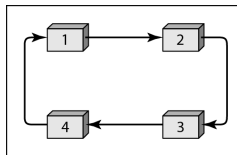
Token-Passing Network

- Transmission right is represented by a special frame: the **token**.
- A station may send data only while holding the token.
- After transmission, token is passed to the next station.
- This enforces orderly, collision-free access.

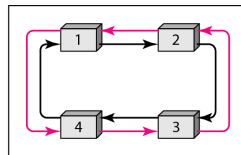


Logical Ring vs Physical Topology

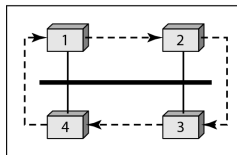
Figure: Token circulation follows a logical order that may differ from physical cabling layout.



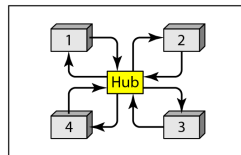
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

12-3 Channelization: Scope

- Channelization shares one link by partitioning resources among stations.
- Partitioning can be by **frequency**, **time**, or **code**.
- Unlike contention methods, structured separation is built in.

Topics in this section

- Frequency-Division Multiple Access (FDMA)
- Time-Division Multiple Access (TDMA)
- Code-Division Multiple Access (CDMA)

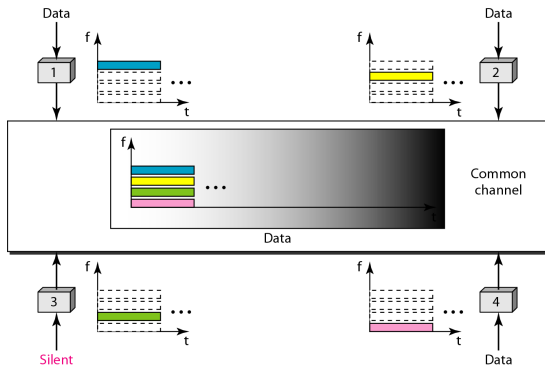
- Total available bandwidth is shared among stations by frequency bands.
- FDMA is a data-link access method that relies on FDM at the physical layer.
- Guard bands separate adjacent user bands to reduce interference.
- Each station keeps its assigned band while active.

Key statement

In FDMA, one common channel is split into frequency bands separated by guard bands.

Frequency-Division Multiple Access

Figure: Users transmit simultaneously in distinct frequency slices of the shared spectrum.



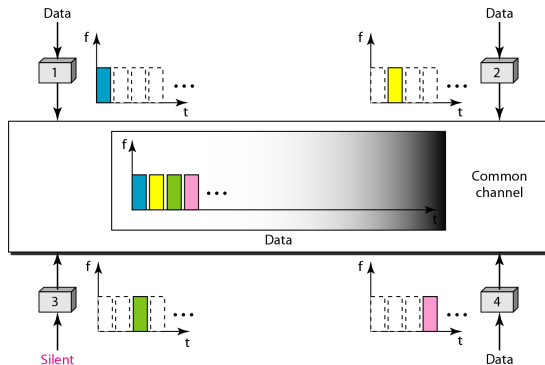
- Entire system bandwidth is treated as one channel.
- Stations share that channel in time.
- Each station transmits in assigned slot(s) within repeating frames.
- Timing alignment is essential for correct slot separation.

Key statement

In TDMA, one channel is time-shared among stations.

Time-Division Multiple Access

Figure: Users reuse the same frequency resource by transmitting in different time slots.



- One channel occupies the entire link bandwidth.
- All stations can transmit simultaneously.
- Separation is achieved by distinct spreading codes, not time or frequency slices.
- CDMA therefore differs fundamentally from both FDMA and TDMA.

Key statement

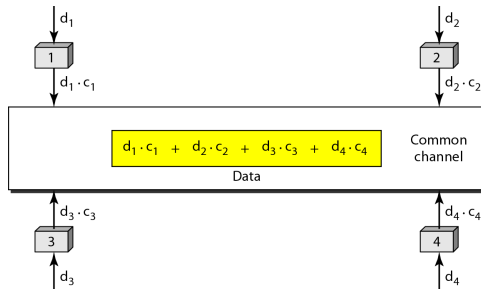
In CDMA, one shared channel carries all transmissions at once using code separation.

Simple Communication Idea in CDMA

- Assigned codes are designed to be orthogonal.
- Cross-correlation of different codes is 0.
- Auto-correlation of a code with itself is N (number of stations/chips).

$$(d_1 c_1 + d_2 c_2 + d_3 c_3 + d_4 c_4) \cdot c_1 = 4d_1$$

Figure: Shows how orthogonality enables one user's recovery from a composite signal.



CDMA Chip Sequences

- CDMA is built on coding theory.
- Each station receives a chip sequence (code).
- Code properties enable simultaneous transmission and separation.

Figure: Provides example chip sequences assigned to stations.



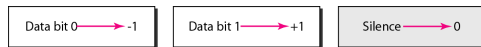
Properties of Orthogonal Sequences

- For identical sequences A : $A \cdot A = N$
- For two different orthogonal sequences A, B : $A \cdot B = 0$
- For sign-inverted sequence: $A \cdot (-A) = -N$
- These identities enable despreading and user separation at the receiver.

CDMA Data Representation

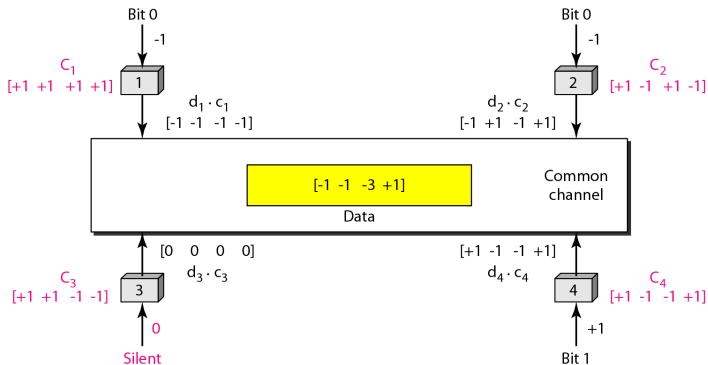
- Data bits are mapped to chip-level signal representations.
- Idle condition is represented by zero signal contribution.
- Active bits are spread according to assigned chip code.

Figure: Links bit values and idle state to CDMA signal representation.



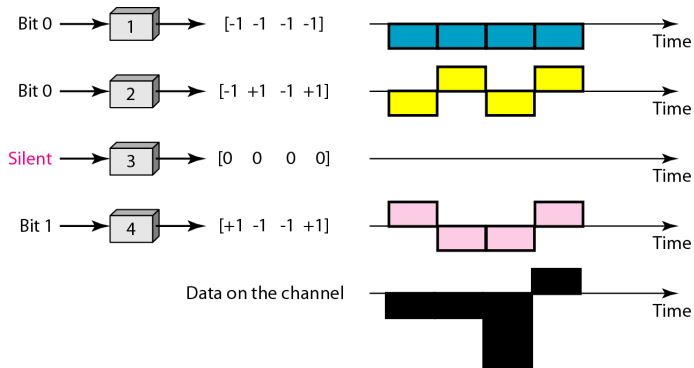
Encoding and Decoding: Sharing a CDMA Channel

Figure: Station signals are encoded, summed, and later decoded using code correlation.



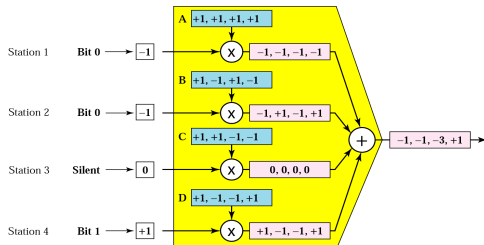
Composite Digital Signal in CDMA

Figure: The transmitted waveform is the superposition of simultaneously spread user signals.



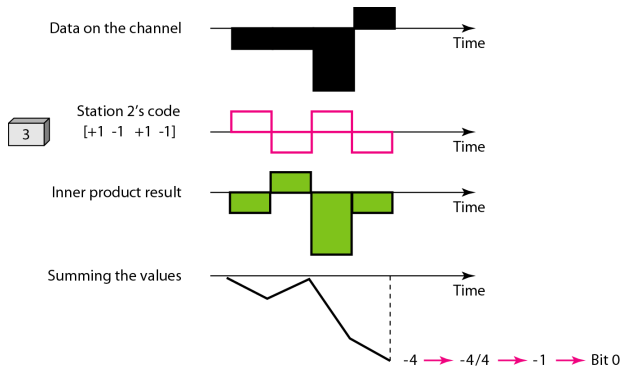
CDMA Encoding Rules: Multiplexer View

- Consider four stations during one bit interval.
- Each station multiplies its bit by its chip sequence.
- The CDMA multiplexer adds all encoded chip streams.
- Result is one composite signal on the shared link.



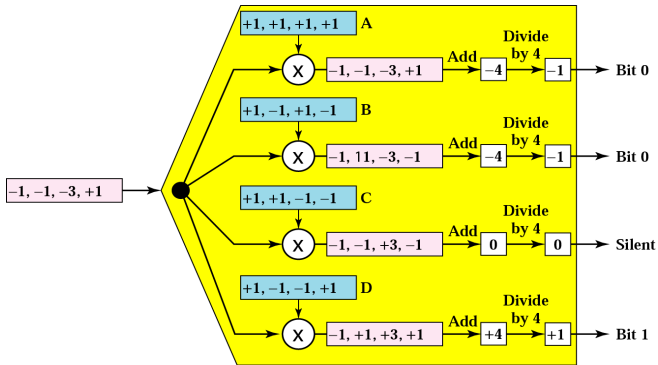
CDMA Decoding Example for One Station

Figure: Correlating the composite signal with one station's code extracts that station's data value.



CDMA Encoding Rules: Demultiplexer View

Figure: The demultiplexer applies user-specific codes to separate parallel data streams from the sum signal.



CDMA Sequence Generation with Walsh Tables

- Walsh tables are square matrices of orthogonal chip sequences.
- Each row can be assigned as a user code.
- Table construction follows recursive orthogonal rules.

Figure: Shows the general Walsh-table construction pattern and examples.

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \qquad W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of W_1 , W_2 , and W_4

CDMA Note: Required Number of Sequences

Note

The number of sequences in a Walsh table must satisfy $N = 2^m$, where m is a nonnegative integer.



Example 12.6: Chips for 2 and 4 Stations

Problem Find chip sequences for:

- (a) Two-station network
- (b) Four-station network

Solution using Walsh rows

- W_2 : $[+1, +1]$, $[+1, -1]$
- W_4 : $[+1, +1, +1, +1]$, $[+1, -1, +1, -1]$, $[+1, +1, -1, -1]$, $[+1, -1, -1, +1]$

Example 12.7: Number of Sequences for 90 Stations

Problem How many orthogonal sequences are needed for 90 stations?

Solution

- Walsh size must be power of two: $N = 2^m$.
- Smallest $2^m \geq 90$ is $2^7 = 128$.
- So choose $m = 7$, generate 128 sequences, and assign 90 of them.

Summary (1)

- Data link layer can be viewed as control sublayer plus shared-medium access sublayer.
- Multiple-access protocols are grouped as random access, controlled access, and channelization.
- In random access, stations contend without fixed authority.
- ALOHA enables decentralized access but is collision-prone.
- Simultaneous transmissions can corrupt frames on the shared medium.

Summary (2)

- CSMA improves performance by sensing before transmitting.
- Persistence methods include 1-persistent, non-persistent, and p -persistent behavior.
- CSMA/CD extends CSMA with collision detection and retransmission logic.
- CSMA/CA targets wireless systems using IFS, contention windows, and ACKs to avoid collisions.

Summary (3)

- In controlled access, stations coordinate authorization before transmission.
- Reservation uses minislots to request future data transmission opportunities.
- Polling relies on a primary controller with poll/select operations.
- Token passing circulates a token in a logical ring to grant send permission.

Summary (4)

- Channelization shares resources by frequency, time, or code.
- FDMA allocates fixed frequency bands.
- TDMA allocates recurring time slots.
- CDMA assigns orthogonal chip sequences and allows simultaneous transmissions.
- Walsh tables provide practical orthogonal sequence generation for CDMA.

Questions and Discussion

- Random vs controlled access tradeoffs
- CSMA/CD and CSMA/CA differences
- FDMA, TDMA, and CDMA comparison

